

Calc 1 Lecture 15:

Hyperbolic functions

Idea: the hyperbolic functions are a "parallel universe" to the trig functions

trig functions

$$\cos(x)$$

$$\sin(x)$$

$$\tan(x)$$

$$\cot(x)$$

$$\sec(x)$$

$$\csc(x)$$

hyperbolic functions

$$\cosh(x)$$

$$\sinh(x)$$

$$\tanh(x)$$

$$\coth(x)$$

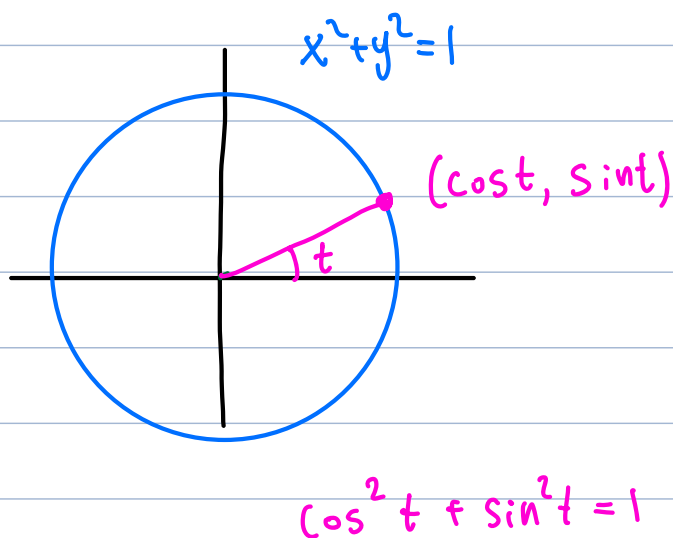
$$\operatorname{sech}(x)$$

$$\operatorname{csch}(x)$$

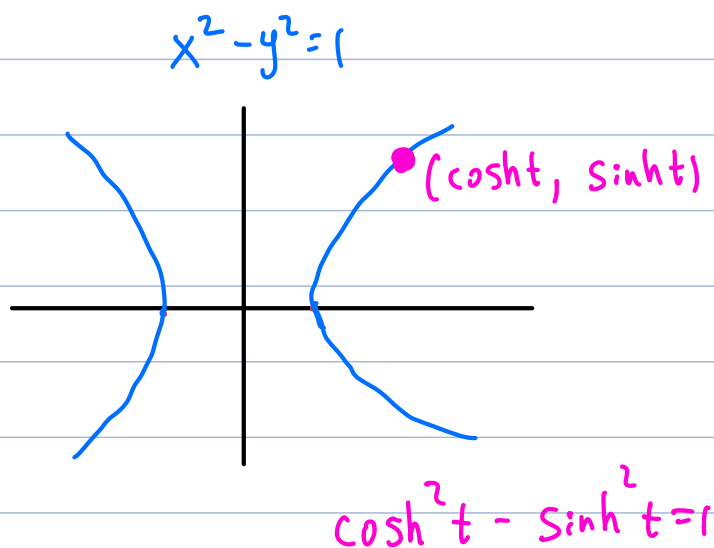
learn
about
these today

The key idea to understanding hyperbolic functions is to replace the "unit circle" with a "unit hyperbola"

Trig world



Hyperbolic world



How to actually define hyperbolic functions?

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$
$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

check the key hyperbolic identity:

$$\cosh^2(t) - \sinh^2(t) = 1$$

$$\begin{aligned}\cosh^2(x) - \sinh^2(x) &= \left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2 \\ &= \frac{(e^x + e^{-x})^2}{4} - \frac{(e^x - e^{-x})^2}{4}\end{aligned}$$

$$(a+b)^2 - (a-b)^2 = (\cancel{a^2} + 2ab + \cancel{b^2}) - (\cancel{a^2} - 2ab + \cancel{b^2}) = 4ab$$

$$\frac{\cancel{4}e^xe^{-x}}{\cancel{4}} = e^xe^{-x} = \frac{e^x}{e^x} = \boxed{1} \quad \checkmark$$

Sign analysis

• when is $\cosh x > 0$? All x .

$$\cosh x = \frac{\overset{\text{Pos}}{e^x} + \overset{\text{Pos}}{e^{-x}}}{2} > 0$$

• when is $\sinh x > 0$?

$\Rightarrow$ when $x > 0$, $\sinh x > 0$
when $x < 0$, $\sinh x < 0$

$$\sinh x = \frac{e^x - e^{-x}}{2} > 0 \stackrel{?}{\Leftrightarrow} e^x > e^{-x}$$
$$\sinh(0) = \frac{e^0 - e^0}{2} = 0 \quad \Leftrightarrow x > 0$$

Also note: $\cosh(x)$ is even ($\cosh(-x) = \cosh(x)$)

$$\text{check: } \cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\cosh(-x) = \frac{e^{-x} + e^x}{2} = \cosh(x) \quad \checkmark$$

$\sinh(x)$ is odd ($\sinh(-x) = -\sinh(x)$)

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$\sinh(-x) = \frac{e^{-x} - e^x}{2} = -\sinh(x)$$

other hyperbolic functions:

the other hyperbolic functions are defined in terms of $\cosh(x)$ and $\sinh(x)$ in familiar ways:

- $\tanh(x) = \frac{\sinh(x)}{\cosh(x)}$
- $\operatorname{sech}(x) = \frac{1}{\cosh(x)}$
- $\coth(x) = \frac{\cosh(x)}{\sinh(x)}$
- $\operatorname{csch}(x) = \frac{1}{\sinh(x)}$

The sech-tanh identity

start with main identity:

$$\cosh^2 x - \sinh^2 x = 1$$

divide
by
 $\cosh^2 x$

$$\frac{\cosh^2 x}{\cosh^2 x} - \frac{\sinh^2 x}{\cosh^2 x} = \frac{1}{\cosh^2 x} \Rightarrow$$

$$1 - \tanh^2(x) = \operatorname{sech}^2(x)$$

Examples: a) Suppose you know $\cosh(x) = \frac{5}{3}$ and $x > 0$.

Find the values of the other hyperbolic functions at x .

b) If $\tanh(x) = \frac{12}{13}$, find the values of the other hyperbolic functions at x .

Solutions: $\cosh x = \boxed{\frac{5}{3}}$

a) idea: use the identity $\cosh^2 x - \sinh^2 x = 1$ to find $\sinh(x)$

$$\sinh^2 x = \cosh^2 x - 1 = \left(\frac{5}{3}\right)^2 - 1 = \frac{25}{9} - 1 = \frac{25-9}{9} = \frac{16}{9}$$

$$\sinh x = \pm \sqrt{\frac{16}{9}} = \pm \frac{4}{3}$$

$$x > 0 \Rightarrow \sinh x > 0 \Rightarrow \sinh x = \boxed{\frac{4}{3}}$$

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{4/3}{5/3} = \boxed{\frac{4}{5}}, \quad \coth x = \boxed{\frac{5}{4}}$$

$$\operatorname{sech} x = \frac{1}{\cosh x} = \boxed{\frac{3}{5}}, \quad \operatorname{csch} x = \frac{1}{\sinh x} = \boxed{\frac{3}{4}}$$

b) If $\tanh(x) = \frac{12}{13}$, find the values of the other hyperbolic functions at x .

$$1 - \tanh^2(x) = \operatorname{sech}^2(x) \Rightarrow \operatorname{sech}^2 x = 1 - \left(\frac{12}{13}\right)^2 = 1 - \frac{144}{169} = \frac{25}{169}$$

$$\Rightarrow \operatorname{sech} x = \pm \sqrt{\frac{25}{169}} = \pm \frac{5}{13}$$

Note: $\cosh x > 0$ for all $x \Rightarrow \operatorname{sech} x = \frac{1}{\cosh x} > 0$

$$\Rightarrow \boxed{\operatorname{sech} x = \frac{5}{13}}$$

$$\boxed{\cosh x = \frac{13}{5}}$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} \Rightarrow \frac{12}{13} = \frac{\sinh(x)}{13/5} \Rightarrow \sinh x = \frac{12}{13} \cdot \frac{13}{5} = \boxed{\frac{12}{5}}$$

$$\boxed{\coth x = \frac{13}{12}}$$

$$\boxed{\operatorname{csc} x = \frac{5}{12}}$$

Derivatives of hyperbolic functions:

$$\begin{aligned}\frac{d}{dx} (\cosh(x)) &= \frac{d}{dx} \left(\frac{e^x + e^{-x}}{2} \right) = \frac{\frac{d}{dx}(e^x) + \frac{d}{dx}(e^{-x})}{2} \\ &= \frac{e^x - e^{-x}}{2} = \boxed{\sinh(x)}\end{aligned}$$

$$\begin{aligned}\frac{d}{dx} (\sinh(x)) &= \frac{d}{dx} \left(\frac{e^x - e^{-x}}{2} \right) = \frac{\frac{d}{dx}(e^x) - \frac{d}{dx}(e^{-x})}{2} \\ &= \frac{e^x + e^{-x}}{2} = \boxed{\cosh(x)}\end{aligned}$$

$$\frac{d}{dx} (\tanh(x)) = \frac{d}{dx} \left(\frac{\sinh x}{\cosh x} \right)$$

$$= \frac{\overbrace{\sinh'(x)}^{\cosh x} \cosh(x) - \sinh(x) \overbrace{\cosh'(x)}^{\sinh x}}{\cosh^2(x)} = \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x}$$

$$= \frac{1}{\cosh^2 x} = \boxed{\operatorname{sech}^2 x}$$

Similarly, can calculate:

$\left. \begin{aligned}\sinh'(x) &= \cosh(x) \\ \cosh'(x) &= \sinh(x) \\ \tanh'(x) &= \operatorname{sech}^2(x)\end{aligned} \right\}$	$\begin{aligned}\operatorname{csch}'(x) &= -\operatorname{csch}(x) \coth(x) \\ \operatorname{sech}'(x) &= -\operatorname{sech}(x) \tanh(x) \\ \coth'(x) &= -\operatorname{csch}^2(x)\end{aligned}$
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Examples:

Find the derivative:

a) $\underbrace{\tanh}_{\text{outside}}(\underbrace{1+e^{2x}}_{\text{inside}})$

$$\frac{d}{dx}(\tanh(1+e^{2x})) = \underbrace{\tanh'(1+e^{2x})}_{\text{sech}^2(1+e^{2x})} \cdot \underbrace{\frac{d}{dx}(1+e^{2x})}_{2e^{2x}}$$

$$= \boxed{2 \text{sech}^2(1+e^{2x}) \cdot e^{2x}}$$

b) $\text{sech}^2(e^t)$

$$\frac{d}{dt}(\text{sech}^2(\underbrace{e^t}_u)) = \frac{d}{dt}(\text{sech}^2(u)) = \underbrace{\frac{d}{du} \text{sech}^2(u)}_{2 \cdot \text{sech}(u) \cdot \frac{d}{du} \text{sech}(u)} \cdot \frac{du}{dt}$$

$$= 2 \text{sech}(u) \cdot (-\text{sech}(u) \tanh(u)) \cdot \underbrace{\frac{du}{dt}}_{e^t}$$

$u = e^t$

$$= -2 \underbrace{\text{sech}(e^t)} \cdot \underbrace{\text{sech}(e^t) \tanh(e^t)} \cdot e^t$$

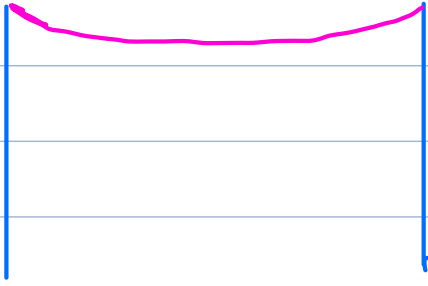
$$= \boxed{-2 \text{sech}^2(e^t) \tanh(e^t) \cdot e^t}$$

Application : "catenary curves"

$$f(x) = c + a \cdot \cosh\left(\frac{x}{a}\right)$$

(a, c are constants)

used to model the shape of hanging wires between poles

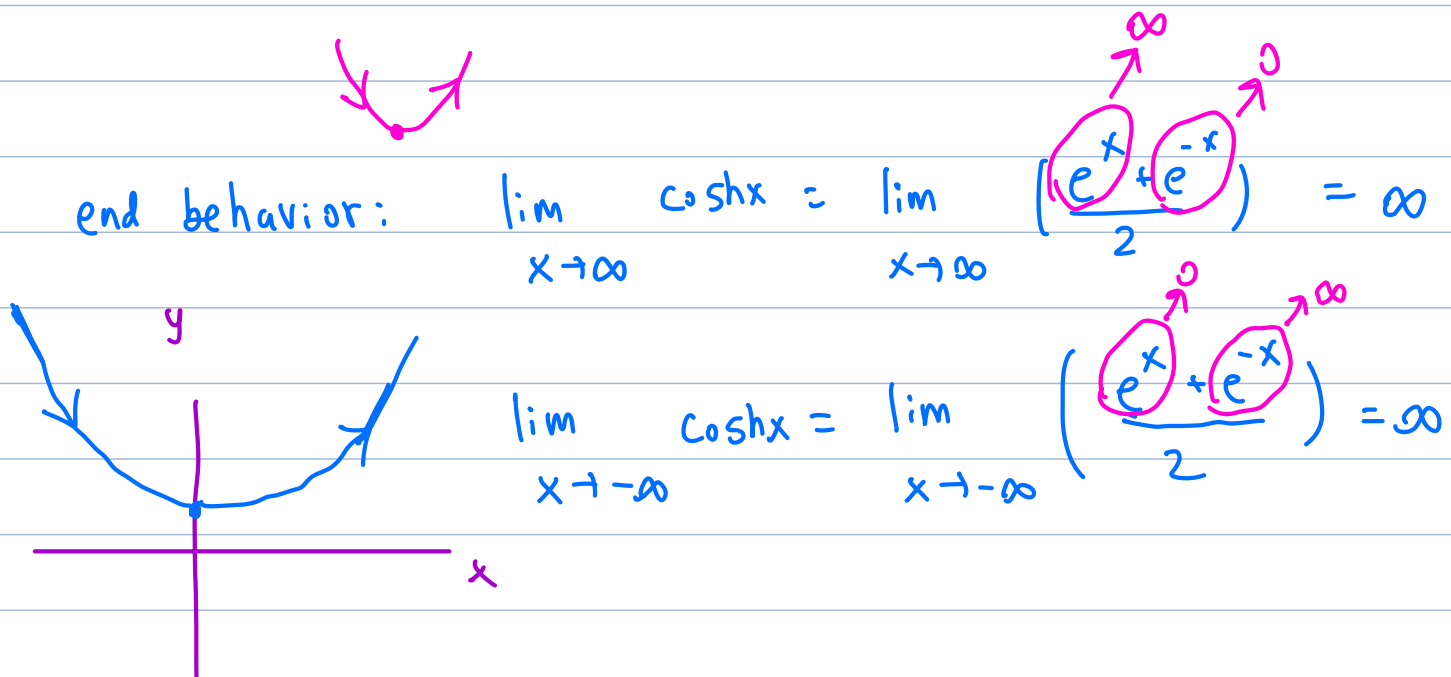


Graphing hyperbolic functions, and their end-behaviors

- $y = \cosh(x)$

$$y' = \sinh(x) \left. \begin{array}{l} \text{Pos. when } x > 0 \\ \text{neg. when } x < 0 \end{array} \right\}$$

$\Rightarrow y = \cosh(x)$ is increasing when $x > 0$
decreasing when $x < 0$

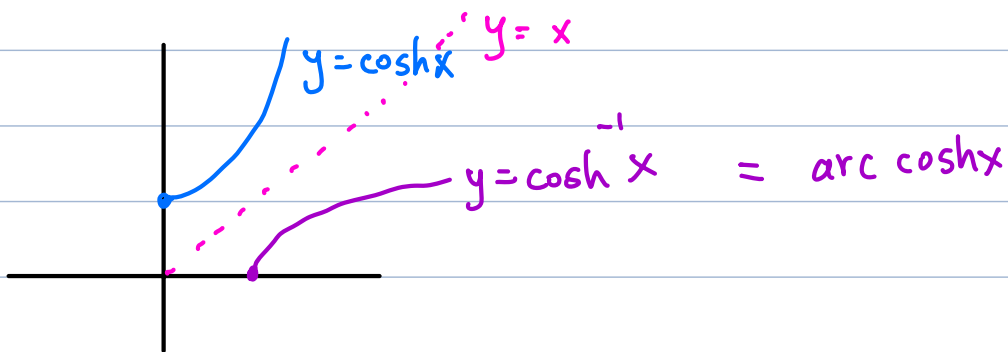


- $y = \sinh(x)$
- $y = \tanh(x)$

Can similarly analyze these using increasing/decreasing info and end behavior

Inverse hyperbolic functions and their derivatives

$y = \cosh^{-1}(x)$ is the inverse function of $\{\cosh x, x \geq 0\}$



$y = \sinh^{-1}(x)$ is the inverse function of $y = \sinh(x)$
(no need to restrict domain here)

Derivatives:

Can use implicit to figure it out.

$$y = \cosh^{-1} x \Rightarrow y' = ?$$



$$\cosh y = x \quad \text{implicit diff}$$

$$\underbrace{\frac{d}{dx}(\cosh y)}_{\sinh y \cdot \frac{dy}{dx}} = \underbrace{\frac{d}{dx}(x)}_1 \Rightarrow \sinh(y) \cdot \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sinh(y)} \Rightarrow \boxed{y' = \frac{1}{\sqrt{x^2 - 1}}}$$

$$\cosh y = x \quad \sinh y = ?$$

$$\underbrace{\cosh^2 y}_{x^2} - \sinh^2 y = 1 \Rightarrow \sinh^2 y = x^2 - 1 \Rightarrow \boxed{\sinh y = \sqrt{x^2 - 1}}$$

Using similar methods, we can find...

$$\frac{d}{dx} (\sinh^{-1}(x)) = \frac{1}{\sqrt{x^2+1}}$$

$$\frac{d}{dx} (\operatorname{csch}^{-1}(x)) = \frac{-1}{|x|\sqrt{x^2+1}}$$

$$\frac{d}{dx} (\cosh^{-1}(x)) = \frac{1}{\sqrt{x^2-1}}$$

$$\frac{d}{dx} (\operatorname{sech}^{-1}(x)) = \frac{-1}{x\sqrt{1-x^2}}$$

$$\frac{d}{dx} (\tanh^{-1}(x)) = \frac{1}{1-x^2}$$

$$\frac{d}{dx} (\operatorname{coth}^{-1}(x)) = \frac{1}{1-x^2}$$

It turns out that we have explicit formulas for the
inverse hyperbolic functions...

$$\sinh^{-1}(x) = \ln(x + \sqrt{x^2+1})$$

$$\cosh^{-1}(x) = \ln(x + \sqrt{x^2-1})$$

} can check using
the definitions